

Soumya Saha

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Experience

Amazon

Seattle, WA

Senior Applied Scientist

Dec 2025 - Present

- Spearheaded central SFT data-mixture optimization and post-training recipe design for Amazon Nova text and multimodal models. Optimized chat templates, training hyperparameters, and efficiency techniques, directly contributing to the successful launch of Nova 2.0.
- Curated high-quality SFT datasets from public and synthetic sources, built automated cleaning, filtering, and model-based scoring pipelines, and provided capability teams with standardized data guidelines and baseline training configurations to improve quality and consistency.
- Designed and executed large-scale systematic ablation studies to refine data composition across diverse domains and modalities, improving downstream model quality while reducing SFT training cycle time by 30%.
- Established model-selection and release-gating workflows spanning capability, safety, and regression evaluations, leading root-cause analysis to trace failures to specific datasets and training configurations, coordinate targeted fixes, and keep high-priority Nova releases on schedule.
- Led SFT, RL, and OPD experiments for customizable open-weight models and advised Nova Forge scientists serving Reddit and MIT Metamaterials on data design, evaluation, and post-training recipes, enabling scalable use-case-specific customization.

Applied Scientist II

Oct 2024 - Nov 2025

- Served as the primary SFT point of contact for Dense (45B to 173B) and MoE (630B-A50B) model releases, coordinating cross-team data contributions, resolving critical regressions under tight timelines, and standardizing data-submission and validation processes.
- Improved key benchmarks (AIME +10%, HMMT +15%, GPQA-Diamond +6%, IFBench +5%) by advancing instruction following, planning, mathematical, and STEM reasoning capabilities in Amazon Nova 1.0 and 2.0 family of models during post-training.
- Collaborated with internal research and annotation teams to collect, evaluate, and iteratively improve high-quality supervised datasets. Conducted systematic ablations on datamix composition and configurations to boost model performance across diverse benchmarks.
- Owned biweekly SFT experiments for the L2 Data Quality program, designing data-composition and training-configuration ablations that improved production model performance and informed subsequent SFT release strategy for any new potential incoming datasets.
- Contributed to the successful launch of Alexa's Language Learning feature by researching and improving text-to-phoneme ASR models for the mispronunciation diagnosis task.

Software Development Engineer

Jul 2020 - Mar 2022

- Integrated Alexa's How To Say Translation feature with Help Me Get Started and migrated the How To Say GUI to APL. Both projects contributed heavily to year-over-year growth, making it Alexa's second-highest-growing feature in 2021.
- Developed and facilitated the release of Alexa's Live Translation feature and expanded it into multiple language pairs.

Software Development Engineer Intern

Jun 2019 - Aug 2019

- Built a self-serve experimental model-building service for the Alexa AI NLU team using AWS, enabling automated training and evaluation of MXNet-based DeepNLU intent-classification and named-entity-recognition models.

Samsung R&D Institute Delhi

Noida, India

Software Development Engineer

Jul 2016 - Jul 2018

- As a C++ developer, led a team of four to build a Native Client (NaCl)-based IPTV reference application with a JavaScript frontend for live and VOD playback on Samsung Tizen Smart TVs, integrating RTP, RTSP, HLS, and Sample AES DRM.

Technical Expertise

Generative AI: LLM post-training (SFT, RLVR, OPD), mathematical reasoning, instruction following, benchmark onboarding, evaluation

Machine Learning: NLP, ASR, speech translation, experimental design, model behavior analysis

Frameworks & Tools: PyTorch, scikit-learn, Kubernetes, AWS, MLflow, Weights & Biases

Education

University of Massachusetts Amherst

Amherst, MA

M.S. Computer Science, Data Science concentration / GPA: 4.0/4.0

Sep 2018 - May 2020

Coursework: NLP, Neural Networks, Machine Learning, Deep Learning for NLP, Reinforcement Learning, Algorithms for Data Science.

Indian Institute of Technology Kharagpur

Kharagpur, India

B.Tech. Electronics and Electrical Communication Engineering / GPA: 8.23/10.0

Jul 2012 - May 2016

Publications

- **The Amazon Nova Family of Models: Technical Report and Model Card** [paper]. Amazon Nova Team. *arXiv preprint*, 2025.
- **FLAMES: Improving LLM Math Reasoning via a Fine-Grained Analysis of the Data Synthesis Pipeline** [paper]. Parker Seegmiller, Kartik Mehta, Soumya Saha, et al. *Findings of EMNLP*, 2025.
- **Improving Low Resource Speech Translation with Data Augmentation and Ensemble Strategies** [paper]. Akshaya Shanbhogue, Soumya Saha, et al. *IWSLT*, 2023.
- **Phonetic RNN-transducer for mispronunciation diagnosis** [paper]. Daniel Yue Zhang, Soumya Saha, et al. *ICASSP*, 2023.
- **Text Embellishment using Attention Based Encoder-Decoder Model** [paper]. Subhajit N, Soumya Saha, et al. *CC-NLG*, 2019.